De La Salle Holy Cross College

Grade 10 Mathematics

Mid-year Examination, Paper 2 – 20 July 2021

Examiner: Mathematics Department Total: 100

Moderators: Grade 10 Mathematics Teachers Time: 2 hours

Instructions and Information:

- This examination consists of 9 pages which include 9 questions. Formulae are provided on page 2.
- Answer every question on folio paper. Number your solutions carefully.
- You may use any permitted calculator. Ensure that it is set in degree mode.
- All necessary working-out must be shown clearly; providing only final answers does not necessarily guarantee full marks.
- Diagrams and graphs are not necessarily drawn to scale.
- Write neatly and legibly; anything that is unclear will not be marked.

	Q 1	Q 2	Q 3	Q 4	Q 5	Q 6	Q 7	Q 8	Q 9	Total
Mark										
Total	9	14	13	12	7	18	12	6	9	100
Marker										

Formulae

Midpoint, M, of line segment AB: $M\left(\frac{x_A+x_B}{2}\ ;\ \frac{y_A+y_B}{2}\right)$

Distance of line segment AB: $AB = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2}$

or $AB^2 = (x_B - x_A)^2 + (y_B - y_A)^2$

Gradient of line segment AB: $m_{AB} = \frac{y_B - y_A}{x_B - x_A}$

Equation of a line: $y = mx + c$ or $y - y_1 = m(x - x_1)$

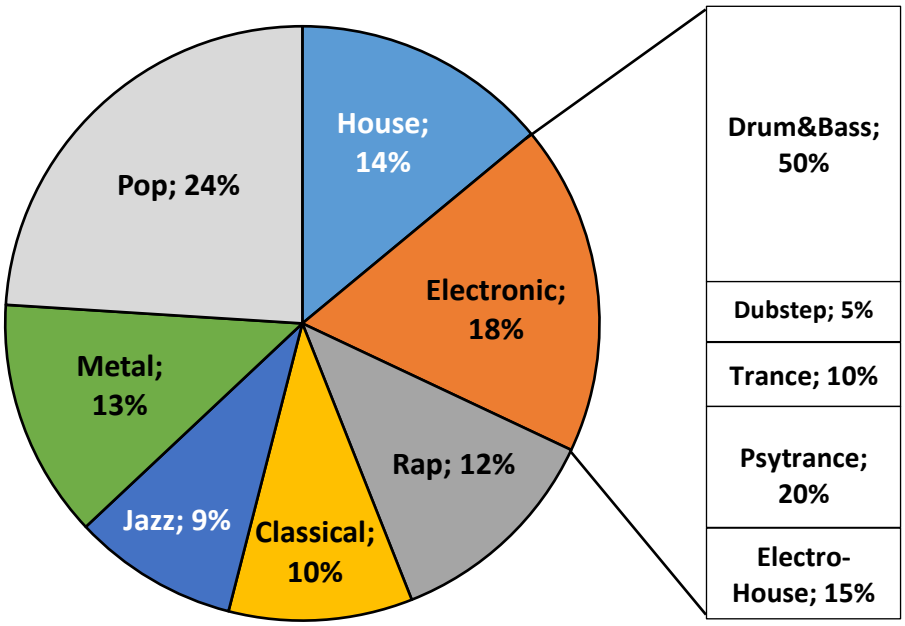
QUESTION 1 [9 Marks]

(a) A survey was conducted in which 50 000 people were asked what their favourite music is. Each respondent could only make their selection from the categories *House, Electronic, Rap, Classical, Jazz, Metal*, or *Pop*.

There was some variation among the fans of electronic music: 50% of them specifically prefer *Drum&Bass*, 5% prefer *Dubstep*, 10% prefer *Trance*, 20% prefer *Psytrance* and the remaining 15% preferred *Electro-House*.

This data was summarised using a pie chart. Use it to answer the questions.

Favourite Music Survey Results of 50 000 people



(1) Determine, without the use of a protractor, the size of the angle representing the *Pop* category correct to the nearest degree. (2)

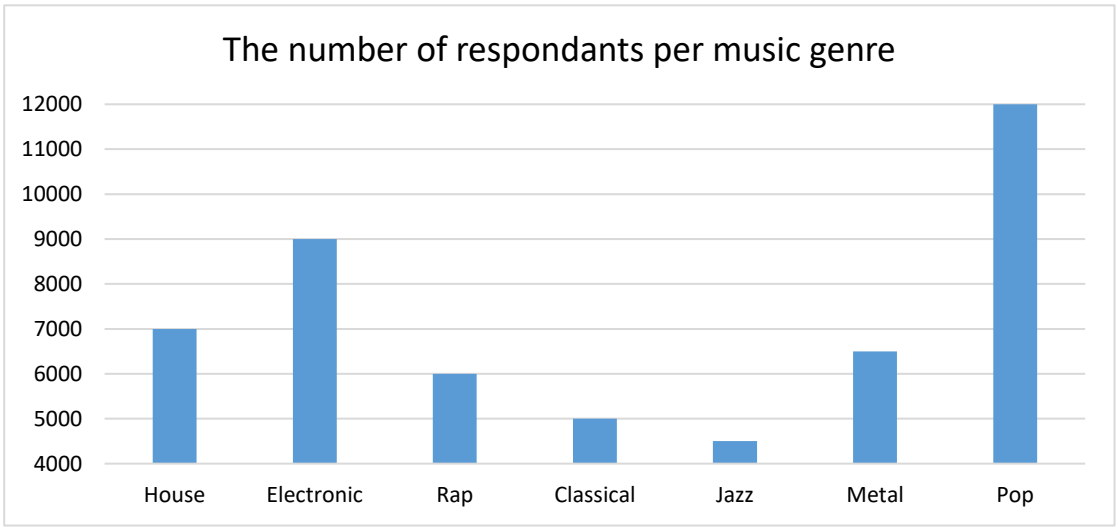
(2) Calculate the **number of participants** who prefer to listen to:

(i) *Metal* music. (1)

(ii) *House* or *Rap* music. (2)

(iii) *Drum&Bass* music. (2)

(b) The manager asked his assistant to represent the number of people who preferred each genre on a bar chart (shown below). (2)

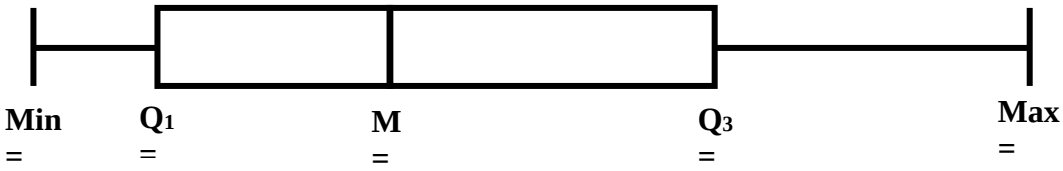


Explain, in words, why the chart is a misleading representation of the data and how this misrepresentation was caused.

QUESTION 2 [14 Marks]

A popular coffee shop advertises its fast service and claims that *most customers are at the coffee shop for 25 minutes*. The manager decides to time how long customers sit at the coffee shop to verify their service claim. The manager summarised the **30** times that he measured on a stem-and-leaf table and then drew a box-and-whisker diagram; both are given below.

STEM	LEAF	ORDERED LEAVES	TOTALS
1/	1 ; 9	1 ; 9	30
2/	7 ; 5 ; 1 ; 5 ; 5 ; 4 ; 2 ; 9	1 ; 2 ; 4 ; 5 ; 5 ; 5 ; 7 ; 9	
3/	8 ; 0	0 ; 8	68
4/	5 ; 6 ; 2 ; 7 ; 9	2 ; 5 ; 6 ; 7 ; 9	
5/	1 ; 7 ; 7	1 ; 7 ; 7	165
6/	2	2	
7/	1 ; 4	1 ; 4	145
8/	2 ; 7	2 ; 7	
9/	3 ; 0 ; 3 ; 8	0 ; 3 ; 3 ; 8	374
10/	2	2	
KEY: 3/8 means 38 minutes			



(a) Find the five-number summary that the manager used to draw the box-and-whisker graph. Fill in your answers on the box-and-whisker graphs above. (5)

(b) Calculate the mean for the data. You should show calculations, but not necessarily the values you used. (3)

(c) The restaurant claims that "*Most customers are at the coffee shop for 25 minutes.*". Name the measure of central tendency that supports this claim and write down this value. (2)

(d) Calculate the **range** and the **IQR** (inter quartile range). (2)

(e) Comment on the distribution of the data and decide if this restaurant provides a reliable service (based on this data). (2)

QUESTION 3 [13 Marks]

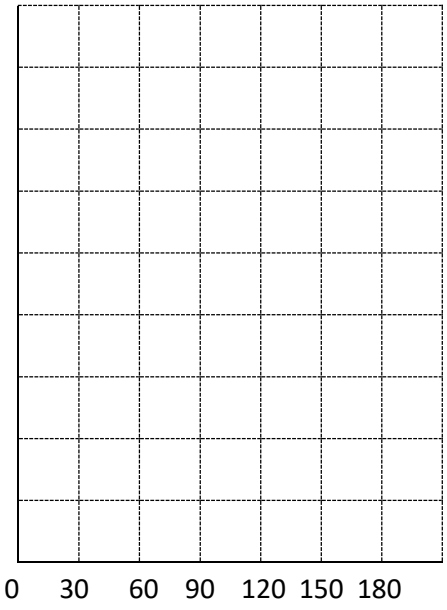
The frequency table below represents the marks (out of a maximum of 180 marks) obtained by a group of Grade 10 pupils in a Mathematics examination.

Mark Interval	Frequency	Class Midpoint	Frequency × Midpoint
$0 \leq m < 30$	6	15	90
$30 \leq m < 60$	12		
$60 \leq m < 90$	38		
$90 \leq m < 120$	42	105	4 410
$120 \leq m < 150$	12		
$150 \leq m \leq 180$	10	165	1 650

(a) Complete the table. (4)

(b) Calculate the estimated mean of the data. (2)

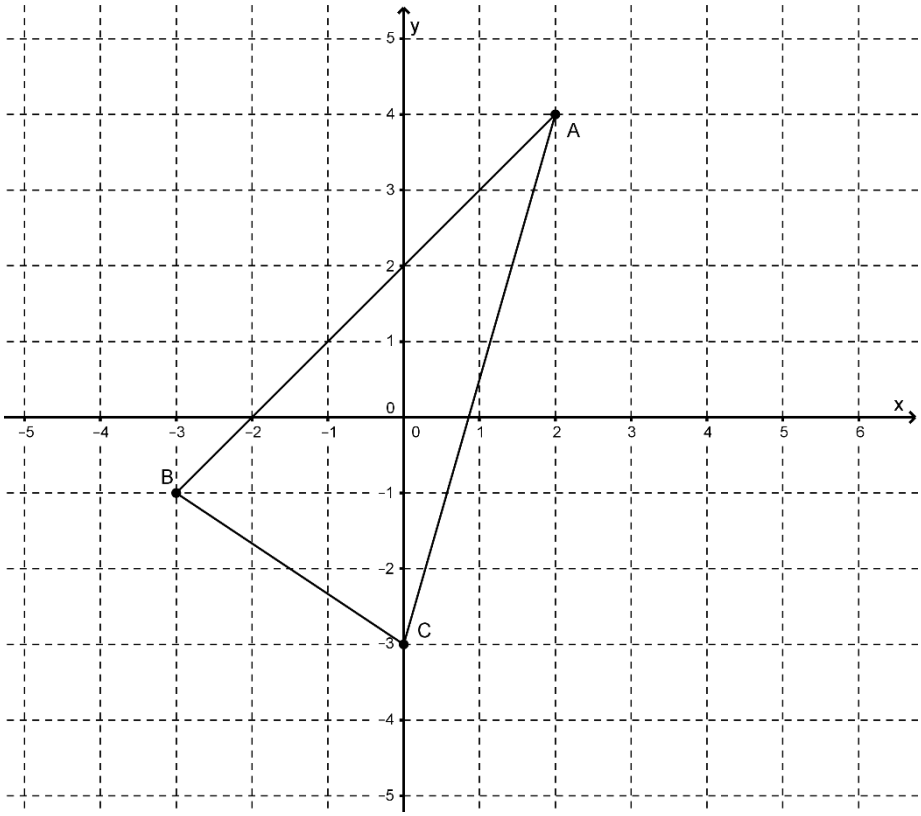
(c) Draw a histogram for the data; give the graph and the axes appropriate headings. Redraw the axes below on your folio paper. (5)



(d) Comment on whether (based on the data) the examination was generally perceived as easy, difficult or fair. Refer to your histogram to substantiate your claims. (2)

QUESTION 4 [12 Marks]

A(2 ; 4),
B(–3 ; –1) and
C(0 ; –3) are the vertices of the triangle shown alongside.



(a) Calculate the distance BC correct to 1 decimal digit. (2)

(b) Determine the gradient of BC. (2)

(c) Hence, determine the equation of the line passing through points B(–3 ; –1) and C(0 ; –3). (2)

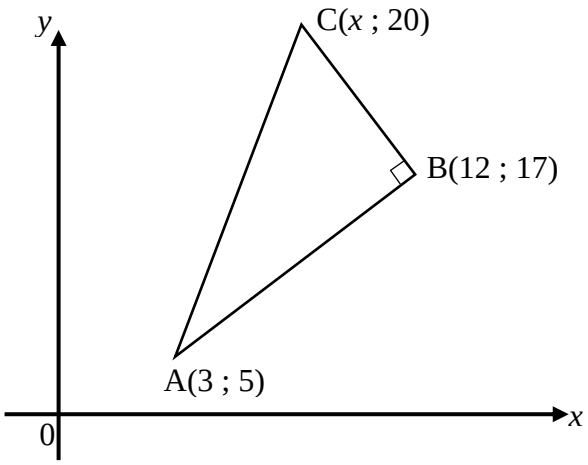
(d) Write down the co-ordinates of D if ADCB is a parallelogram, and both the *x* and *y* co-ordinates of D are positive. (2)

(e) If A(2 ; 4), B(–3 ; –1) and E(7 ; *p*) are three collinear points, then calculate *p*. (4)

QUESTION 5 [7 Marks]

In the diagram alongside ∆ABC is a right-angle scalene triangle with CB⊥AB.

The vertices of △ABC are A(3 ; 5), B(12 ; 17) and C(x ; 20) in the Cartesian plane.



(a) Determine the gradient of AB. (2)

(b) Hence, write down the gradient of BC. (2)

(c) Now, determine the value of x. (3)

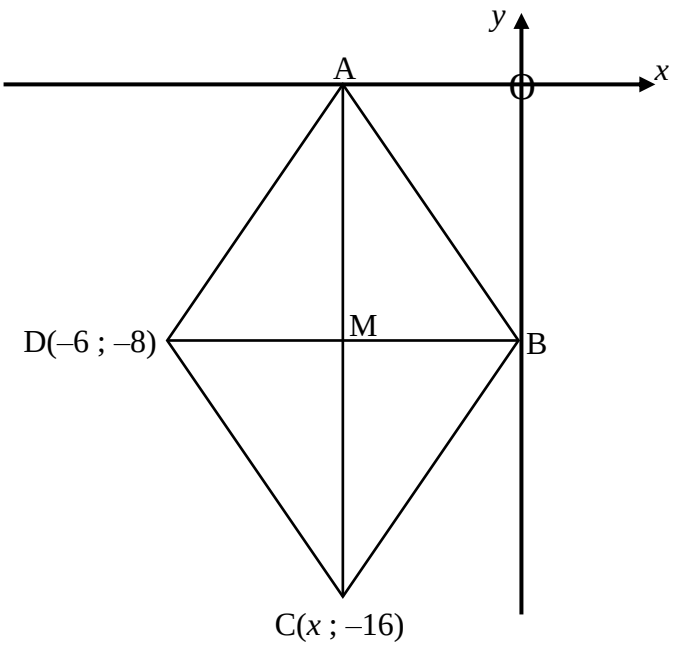
QUESTION 6 [18 Marks]

ABCD is a kite with A and B on the *x*-axis and *y*-axis respectively.

AD = AB and DB is parallel to the *x*-axis.

M is the point of intersection of BD and AC.

D(–6 ; –8) and C(x ; –16)



(a) Write down the co-ordinates of points **B and M**. (4)

(b) Determine the co-ordinates of points **A and C**. (3)

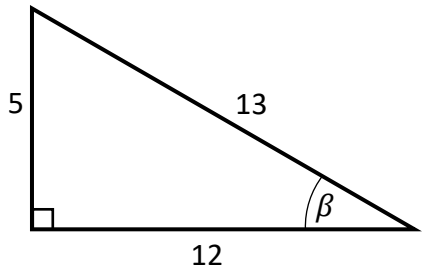
(c) Calculate the area of △AOB (3)

(d) Prove that ABCD is a rhombus, and give the reason why this is so. (5)

(e) Find the equation of the line that is parallel to AB and which goes through the point (5 ; –2). (3)

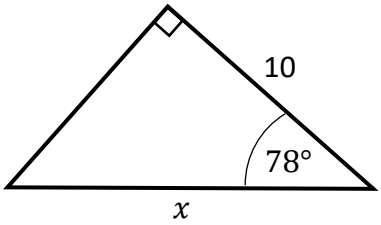
QUESTION 7 [12 Marks]

(a) **[You should not need to use a calculator for this question.]** (4)

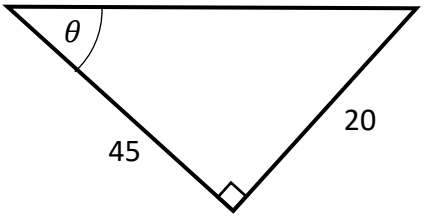


Using the diagram above, arrange the values $\cot\beta$, $\tan\beta$ and $\cos\beta$ in ascending order.

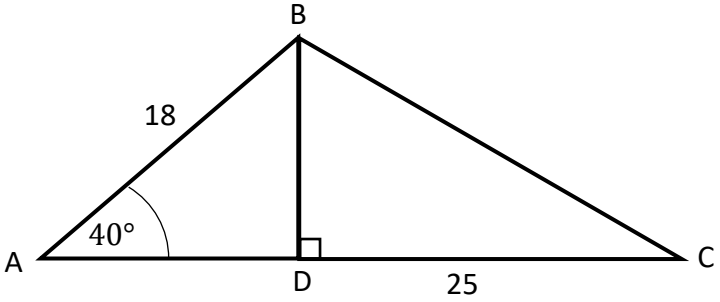
(b) Calculate the length x correct to two decimal digits. (2)



(c) Calculate the size of θ correct to two decimal digits. (2)



(d) In the diagram, AB = 18 units, DC = 25 units and $\hat{A}$ = 40°. Calculate the length of BC correct to the nearest unit. (4)



QUESTION 8 [6 Marks]

Solve each equation for $x \in [0° ; 90°]$. Round off your answers to the nearest degree.

(a) $12 \tan x - 30 = 0$ (2)

(b) $\sin x = \tan 45°$ (2)

(c) $\cos(2x + 10°) = 0{,}5$ (2)

QUESTION 9 [9 Marks]

(a) Without a calculator, evaluate $40. \cos 30° . \tan 30° . \sin^2 30°$ (5)

(b) While chatting to a matric pupil, Miquela was told of a formula for determining the tangent of the sum of two angles: (4)

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha . \tan \beta}$$

If it is given that $\tan 10° = 0{,}5$, then use the above formula and special angles to determine the value of $\tan 55°$.